Prediction of Vertical Profile of NO_2 Using Deep Multimodal Fusion Network Based on the Ground-Based 3-D Remote Sensing

Shulin Zhang, Bo Li, Lei Liu, Qihou Hu, Haoran Liu, Rui Zheng, Yizhi Zhu, Ting Liu, Mingzhai Sun, and Cheng Liu

***Abstract*— The vertical distribution profiles of NO_2 are essential for understanding the mechanisms, detecting near-surface emissions, and tracking pollutant transportation at high altitude. However, most of the published NO_2 studies are based on the surface 2-D measurements. The ground-based 3-D remote-sensing stations were recently built to measure vertical distribution profiles of NO_2. However, the stations were spatially sparse due to the high cost and could not make the measurements without sunlight. In this study, we first developed a multimodel fusion network (MF-net) based on the sparse vertical observations from the Jing-Jin-Ji region. We achieved the 3-D profile prediction of NO_2 in the range of 39.005–41.405N and 115.005–117.905E with 24-h coverage. The MF-net significantly surpassed the conventional WRF-CHEM model and provided a more accurate evaluation of the NO_2 transmission between Beijing and the neighboring cities. Besides, the MF-net covers the monitoring of NO_2 to the whole study area and extends the monitoring time to the entire day (24 h), making it serviceable for continuous spatial-temporal estimation of NO_2 and its transmission in pollution events. The MF-net provides more robust data support to formulate reasonable and effective pollution prevention and control measures.**

***Index Terms*— 3-D prediction of NO_2, deep learning neural network, multiaxis differential optical absorption spectroscopy (MAX-DOAS), multimodal information fusion, remote sensing.**

I. Introduction

In recent years, people have increasingly been concerned about air pollution with the development of urban industry and transportation. Nitrogen dioxide (NO_2), as one of the major pollutants in the atmosphere, is toxic to human health and causes severe damage to the immune and respiratory systems. The long-term exposure to NO_2 can increase the risk of all-cause mortality [1], even hurt the plant [2]. Moreover, NO_2 is the precursor of secondary aerosols nitrate and has a significant impact on the formation and destruction of other air pollutants. It catalyzes the production of ozone (O_3) in the troposphere and the removal of O_3 in the stratospheric. Therefore, NO_2 plays a vital role in both tropospheric and stratospheric chemistry. Thus, the prevention and control of NO_2 pollution are essential.

Accurate measurement of the NO_2 concentration is the premise to prevent and control of NO_2 pollution, as well as to study the pollution mechanism. Most of the published NO_2 studies are based on the surface measurements of NO_2, such as the historical observations of the China National Environmental Monitoring Center (CNEMC). However, surface NO_2 measurement is not sufficient to fully characterize the pollution, since the pollution is caused not only by the local anthropogenic emissions (industrial emissions, power generation, and traffic emissions), but also remote transportation [3] that happens at high altitudes.

To this end, a type of passive remote-sensing measurement system called multiaxis differential optical absorption spectroscopy (MAX-DOAS) has been proposed [4]. And it has been widely used in the past decade to measure the vertical structure of tropospheric aerosol [5]–[7] and trace gases [8]–[10] by applying the differential optical absorption spectroscopy (DOAS) method to the observations in several different viewing directions. The spatial distribution data of the observations are retrieved from the molecular absorption in the ultraviolet (UV) and visible (VIS) wavelength bands measured from the scattered sunlight spectra. In this study, the dark current and offset spectra were used to correct measurement spectra before the spectra analysis. And we analyzed the MAX-DOAS spectra by the QDOAS spectral fitting software suite developed by BIRA-IASB (http://uv-vis.aeronomie.be/software/QDOAS/). The wavelength was calibrated with a high-resolution solar reference spectrum [11]. The DOAS fit included several trace gas absorption cross sections [12]–[15], the Ring spectrum, a Frauenhofer reference spectrum, and a low-order polynomial. The details can be found in our previous work [3], [16]. Currently, the available column densities obtained from MAX-DOAS includes the atmospheric trace gases (e.g., nitrogen oxides (NOx) [3], [17], sulfur dioxide (SO_2) [18], formaldehyde (HCHO) [19],

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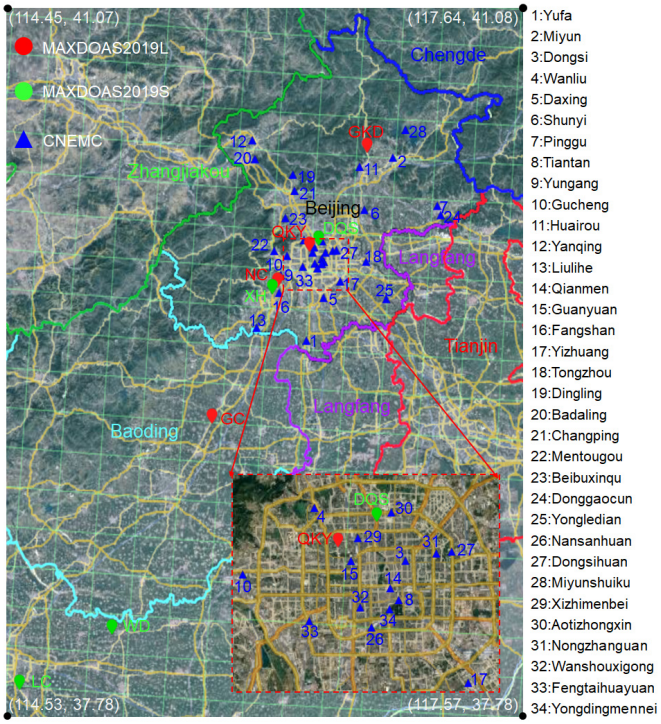


Fig. 1. Location of the MAX-DOAS instruments and the CNEMC sites. The red dots represent the four MAX-DOAS stations with longer term measurements, while the greens marked the four MAX-DOAS sites containing shorter term observations, and the blue triangles are 34 Beijing CNEMC sites. The green line represented the grid of WRF-CHEM with a spatial resolution of 20 × 20 km.

Oxygen dimer (O_4) [20], [21], Nitrous acid (HONO) [9], Glyoxal [22]), and tropospheric column densities, surface volume mixing ratios (VMRs), vertical profiles of aerosol extinction, and trace gas mixing ratios. The MAX-DOAS observations have been performed on a range of flexible applications, such as ground-based stations [16] and mobile platforms (e.g., cars [23], ships [19], and aircraft [24]). However, the observation range of ground-based stations is limited in space and time. It can only observe the vertical profile of NO_2 at the observation site's location during the day and is interrupted at night. To cover the entire urban area needs a large number of observation sites. However, it is impracticable to build a large amount of MAX-DOAS sites due to the high cost of building and maintenance.

In this study, we collected the historical data from four MAX-DOAS stations in Beijing-Tianjin-Hebei (Jing-Jin-Ji) region (Fig. 1), which was especially severely polluted due to its intensive use of fossil fuel for dense population's energy and transportation demand [7]. We proposed a NO_2 3-D prediction multimodal fusion network (MF-net) that could fully use the rich information from different types of data, including the simulation results of NO_2 and the meteorological parameters from WRF-CHEM. After training with the vertical profile of NO_2 from MAX-DOAS measurement, the network shows superior performance than WRF-CHEM simulation in all experimental results. The main contributions of the study are as follows.

1) We achieved 3-D prediction of the vertical profile of NO_2 distribution using the proposed neural network (MF-net) trained on the observations. The MF-net was able to cover the entire study area and extend the monitoring time from daytime to the whole day, including night.
2) With the MF-net, we were able to evaluate the 3-D NO_2 transmission between Beijing and the neighboring cities more accurately from October 2016 to November 2019.
3) We applied the MF-net to study the dynamics of a pollution event with a much higher spatial-temporal continuity, which help understand the mechanism of NO_2 pollution in Beijing and formulating air quality assurance policies.

This article was organized as follows. In Section II, we introduced related researches about air pollutant prediction. In Section III, we described the details about the data we used in our study. The proposed 3-D prediction approach of MF-net was also introduced in this section, as well as the experimental process of the training and testing. Next, we first compared the performance of MF-net and WRF-CHEM in four experiments in Section IV-A. Then we demonstrated the application value of MF-net by applying the algorithm in three practical applications in Section IV-B. The discussion and conclusion of this work were in Sections V and VI.

II. Related Work

Current 3-D prediction of air pollution can be grouped into two categories: simulation models and neural network-based models. To simulate both meteorological component and chemical processes, Byun [25] proposed a computationally attractive approach [the community model for air quality (CMAQ)] for it produced many chemical transport simulations with only a single meteorological data set in the retrospective offline chemical transport simulations. To avoid failing in capturing the atmospheric processes with a less time scale than the meteorological model, typically available once or twice per hour, the weather research and forecasting (WRF) model (http://www.mmm.ucar.edu/wrf/users/document.html) was developed with the applicability on many different scales for the dynamic cores and different choices for physical parameterizations. Bacon *et al.* [26] applied the unstructured grid to implement a continuously varying horizontal and vertical grid resolution. And Satoh [27] finished a similar work in the Japanese numerical weather prediction model. Grell *et al.* [28], [29] implemented various chemical modules into the WRF model in the study MM5/Chem and WRF-CHEM. In the WRF-CHEM model, both the meteorology and chemistry were fully coupled "online," with higher temporal and spatial resolution.

Deep learning methods have been widely used in solving many complex problems, including atmospheric prediction. Zheng *et al.* [30] proposed a semisupervised deep learning approach to infer the real-time and fine-grained air quality information based on the air quality data and a variety of data sources (such as meteorology, traffic flow, human mobility,

the structure of road networks, and point of interests (POI). Zheng *et al.* [31] also developed a predictive model providing 48-h fine-grained air quality forecasts for four major Chinese cities every hour. Qi *et al.* [32] proposed a deep learning network called deep air learning (DAL) to solve the interpolation, prediction, and feature analysis problems in one model. To estimate $PM_{2.5}$, deep belief network and CNN-LSTM were applied with considering spatiotemporally correlation [33], [34]. Cabaneros *et al.* [35] and Valput *et al.* [36] achieved forecasting NO_2 concentration by neural networks.

Currently, there is no deep learning research based on the sparse MAX-DOAS observations for several reasons. First, the observation data are minimal because there are very few stations, and most stations are constructed in recent years. Besides, the MAX-DOAS instruments can only measure NO_2 during the daytime, not at night. Second, the unfavorable weather conditions interrupt the measurement process intermittently and cause substantial measurement errors or missing data. Third, the NO_2 prediction is very challenging since it involves both meteorological factors (such as wind speed and direction, turbulence, radiation, clouds, and precipitation) and chemical processes (such as deposition and transformations), as well as transport contribution. Fourth, the stations are so sparse that it makes the interpolation very difficult for most of the grid without historical data.

Evident from the situations above, there are currently no deep-learning-based studies on 3-D prediction of NO_2 using MAX-DOAS observations. To this end, we proposed a multimodal fusion network (MF-net), which entirely uses the vertical profile information of NO_2 from MAX-DOAS. The MF-net fused the multimodal data, including meteorological data and vertical concentration of NO_2 from WRF-CHEM simulation, and achieved the 3-D prediction of NO_2 at any grid point with or without historical data. The stereoscopic monitoring of NO_2 in an area provides valuable information for locating the sources of local emission and transmission. To compare the performance of the MF-net and the WRF-CHEM model, we systemically evaluated both methods on MAX-DOAS measurements and other testing data sets, such as the 2-D NO_2 surface measurement from the CNEMC sites. All experimental results show that the MF-net significantly outperforms WRF-CHEM in NO_2 prediction.

III. Methodology

A. Data Sets

In this section, we introduce all the data sets we applied in the study.

1) Data Sets of the MAX-DOAS Observations: Our MAX-DOAS instruments provided the measurements and errors of NO_2 concentration at daytime with a measurement intermittent of 15 min. The observations contained 16 vertical altitudes, from the surface layer to 3 km with an interval of 200 m. However, due to the high miss rate and large measurement errors above 1 km, only the measurements of six altitude layers within 0∼1 km are used in our study. The details of the way to obtain the NO_2 vertical profiles from the MAX-DOAS instruments were similar to our previous work [3], [16]. We separated the observations of MAX-DOAS stations into two data sets by whether the period of historical measurement was more than one year. The MAX-DOAS-L data set described in Table I consisted of four stations with longer term MAX-DOAS observations while specifying the time range of one to three years. It included Gucheng (GC), Guokeda (GKD), Nancheng (NC), and Qikeyuan (QKY), with 4672, 512, 3458, and 7076 samples, respectively. The MAX-DOAS-S (Table II) were acquired from stations at DQS, Wangdu (WD), Xianghe (XH), and Luancheng (LC) in a shorter period. The longitude, latitude, and the observation period of each MAX-DOAS station were listed in Tables I and II. In this study, MAX-DOAS-L was using for training and cross-validation, whereas MAX-DOAS-S was only used for validation.

TABLE I
DETAILS OF THE MAX-DOAS-L DATA SET

Station	Longitude	Latitude	Period
GC[1]	115.734°	39.149°	2016/10/21 ∼ 2019/03/24
GKD[2]	116.675°	40.408°	2018/02/26 ∼ 2019/12/31
NC[3]	116.127°	39.781°	2018/04/27 ∼ 2019/12/31
QKY[4]	116.3206°	39.9472°	2016/10/20 ∼ 2019/12/21

[1]Gucheng, [2]Guokeda, [3]Nancheng, [4]Qikeyuan.

TABLE II
DETAILS OF THE MAX-DOAS-S DATA SET

Station	Longitude	Latitude	Period
DQS[1]	116.3761°	39.975°	2019/09/21 ∼ 2019/11/30
WD[2]	115.15°	38.17°	2019/04/01 ∼ 2019/11/30
XH[3]	116.095°	39.75°	2018/12/03 ∼ 2019/11/30
LC[4]	114.6097°	37.9072°	2019/10/11 ∼ 2019/11/30

[1]The Institute of Atmospheric Physics, [2]Wangdu, [3]Xianghe, [4]Luancheng.

2) Meteorological Data: The 3-D distribution of the temperature, pressure, longitude wind speed, latitude wind speed, vertical wind speed, and the atmospheric pollutant of NO_2 for the input of the MF-net was simulated using the WRF-CHEM model [3] with version 3.7.1. The model domain was configured to cover the Jing-Jin-Ji and its surrounding regions with a grid resolution of 20 × 20 km and 44 vertical layers. To initialize the boundary conditions of meteorological fields for the WRF-CHEM simulation, we adopted the 6-h final operational global analysis data generated from the National Center for Environmental Prediction (NCEP) with a spatial resolution of 1° × 1°. The inputs of the emission inventory were obtained from the Multiresolution Emission Inventory for China (MEIC, http://www.meicmodel.org/) [37], [38]. The simulations were conducted month by month and the outputs from previous runs were used as the initial conditions of the following run. To minimize the influence of initial conditions, the first five days of each month were simulated and discarded as a spin-up period. Details of model configuration options can be found in our previous studies [1], [39].

3) Surface Observations: The surface NO_2 observation data, which we used to validate the MF-net, were

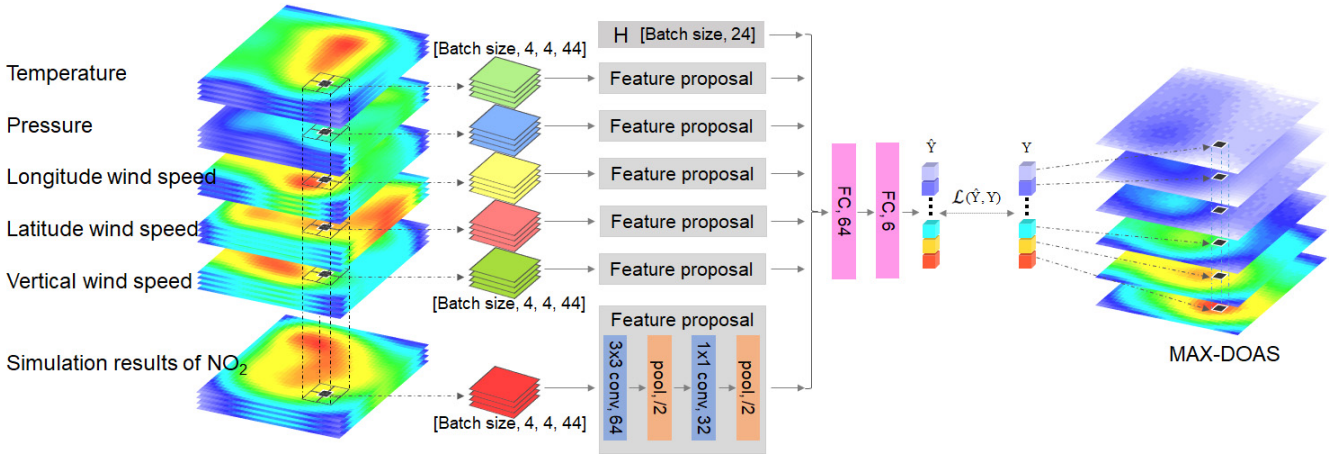


Fig. 2. Overview structure of the MF-net. Each color patch (second column) corresponded to one of the six types of MF-net inputs, denoting with [Batch size, 4, 4, 44], showing that the patch size was 4×4 with 44 vertical layers. The batch size was the number of samples used to update the MF-net weights in each iteration. Each gray box (third column) represented a feature extraction module and had the same structure consisting of two convolutional layers followed by max pooling operation. The kernel size and output channels of the convolutional layer were provided inside of the blue boxes. The notation of "pool, /2" represented a pooling layer with a stride of 2×2. The number in the magenta boxes marked the node number of the fully connection layer.

collected by 34 CNEMC stations located in Beijing (http://beijingair.sinaapp.com/). The concentration of the surface air pollutants including air quality index (AQI), $PM_{2.5}$, NO_2, and SO_2 was measured every hour between 2018 and 2019.

4) Satellite Observations: The tropospheric vertical column densities (VCDs) of NO_2 were retrieved from the TROPOspheric Monitoring Instrument (TROPOMI) instrument based on the algorithm of Zhang *et al.* [40], [41]. In our previous work [41], we regridded the tropospheric NO_2 VCDs per orbit to the resolution of $0.1^\circ \times 0.1^\circ$ over eastern China using a novel P-Spline method [42]. There was only one measurement result per day in Beijing and its surrounding areas, and the corresponding observation time is 3:30 (UTC).

B. MF-Net Architecture

The architecture of the MF-net is shown in Fig. 2. The MF-net started with six feature extraction modules with the same architecture consisted of two convolutional layers followed by max pooling operation. It ended by two fully connection layers fused the features obtained from the six feature extraction modules and H. H represented a 1-D time vector, which can be expressed as $(t_1, t_2, t_3, \ldots, t_{24})$ where t_i equals 1 when the hour of the sample date was i, otherwise 0. As shown in Fig. 2, every input of MF-net was a 4×4 patch with 44 layers, which was cropped from the meteorological data (such as temperature, pressure, and the wind speed of latitude, longitude, and vertical) and the vertical NO_2 distribution from the WRF-CHEM simulation centering at the target point. The predictions of MF-net were the vertical distribution of NO_2 concentrations at six attitudes, from the surface to 1 km with an interval of 200 m, at the target points. The loss function of the MF-net is defined as

$$\mathcal{L}(Y, \hat{Y}) = \log(\cosh(Y, \hat{Y})) \tag{1}$$

where $\hat{Y}$ and Y are the prediction and the corresponding MAX-DOAS observation of NO_2 concentration at the six vertical altitudes. As the input of MF-net is based on the simulation of WRF-CHEM, the spatial resolution of MF-net is 20×20 km and the temporal resolution is 1 h.

C. Training and Testing MF-Net

In this study, to improve the stability of the algorithm, we removed the observations of MAX-DOAS with large measurement errors, including the root mean square (RMS) during the DOAS fit was larger than 0.1 or the corresponding aerosol optical depth was larger than 3. Then, we interpolated the cleaned MAX-DOAS observations to the integer hour. Specifically, the average of the observations during the half hour before and after each hour is used as the measurement result at the integer hour.

We applied the fourfold cross-validation on the MAX-DOAS-L data set. Specifically, the observations were split into four subsets by site. To ensure complete independence of the subsets data, observations from the same site were assigned into the same subsets. Then, we performed four experiments to ensure that each subset was used as a test set and the remaining subsets as the training set.

To further test the performance of the MF-net, we first trained the MF-net with all of the observations in the MAX-DOAS-L data set. Then we evaluated the trained MF-net by testing on the other three data sets: the MAX-DOAS-S, CNEMC, and the satellite data set, respectively. When testing on the observations of 34 CNEMC stations, we applied two evaluation conditions. In the first experiment, we evaluated the surface prediction of MF-net on the CNEMC observations that were most accurately monitored from 21:00 to 11:00 next day of UTC hour. For the second experiment, we analyzed the performance of MF-net for whole day observations.

The MF-net was implemented with TensorFlow using Python 3.6. The experiments were performed on a cluster with ten GeForce GTX 1080Ti GPUs. All the convolutional layers were initialized randomly, and the biases were initialized

with 0. During training, we applied data augmentation by adding random Gaussian noise $[\mathcal{N}(0, 0.005)]$ to the inputs. Besides, all the inputs were normalized by subtracting the mean pixel value of each channel of the image followed by dividing the corresponding standard deviation. The MF-net was trained for 100 epochs with batch size of 32, an Adam optimizer with β_1 of 0.9, β_2 of 0.999, and a learning rate of 0.0005. On average, it took 596 s to train the MF-net with all of the observations in the MAX-DOAS-L data set on a GeForce GTX 1080Ti GPU. To process one sample, the MF-net took about 0.22 ms.

D. Evaluation Metrics

To evaluate the performance of our MF-net, we use the Pearson correlation as the evaluation metric. Specifically, the observations of NO_2 vertical concentration are noted as $Y \in \{\boldsymbol{y}_i; i = 1, 2, \ldots, n\}$ and $\hat{Y} \in \{\hat{\boldsymbol{y}}_i; i = 1, 2, \ldots, n\}$ is the corresponding prediction where n is the number of the observations. The ith observation is represented as $\boldsymbol{y}_i = (y_i^{(1)}, y_i^{(2)}, y_i^{(3)}, y_i^{(4)}, y_i^{(5)}, y_i^{(6)})$ for containing six vertical altitudes. Similarly, we use $\hat{\boldsymbol{y}}_i = (\hat{y}_i^{(1)}, \hat{y}_i^{(2)}, \hat{y}_i^{(3)}, \hat{y}_i^{(4)}, \hat{y}_i^{(5)}, \hat{y}_i^{(6)})$ for the response predictions. The P_h represents the Pearson correlation between the observations and the predictions of NO_2 in h_{th} vertical altitudes is defined as

$$P_h = \text{Corr}\left(Y^{(h)}, \hat{Y}^{(h)}\right) = \frac{\sum_{i=1}^{n}\left(y_i^{(h)} - \overline{Y^{(h)}}\right) \times \left(\hat{y}_i^{(h)} - \overline{\hat{Y}^{(h)}}\right)}{\sqrt{\sum_{i=1}^{n}\left(y_i^{(h)} - \overline{Y^{(h)}}\right)^2} \times \sqrt{\sum_{i=1}^{n}\left(\hat{y}_i^{(h)} - \overline{\hat{Y}^{(h)}}\right)^2}} \quad h \in \{1, 2, 3, 4, 5, 6\} \tag{2}$$

where $Y^{(h)} \in \{y_i^{(h)}; i = 1, 2, \ldots, n\}$ and $\hat{Y}^{(h)} \in \{\hat{y}_i^{(h)}; i = 1, 2, \ldots, n\}$ represent the observations and the predictions of the NO_2 concentration at h_{th} vertical altitude, respectively. We used $P_{\text{avg}} = \frac{1}{6}\sum_{h=1}^{6} P_h$ to represent the average Pearson correlation between the observations and the predictions at six altitudes. Since the previous evaluation indicators focused on 2-D prediction, which is not conducive to the comparison of the model prediction in 3-D, we proposed P_{3-D} as the comprehensive indicator to evaluate the model performance in six altitudes. The P_{3-D} was a more appropriate evaluation metric for the distribution of concentration along the vertical direction

$$P_{3D} = \mathbb{E}\{\text{Corr}\left(Y, \hat{Y}\right)\} = \frac{1}{n}\sum_{i=1}^{n} \frac{\sum_{h=1}^{6}\left(y_i^{(h)} - \overline{Y_i}\right) \times \left(\hat{y}_i^{(h)} - \overline{\hat{Y}_i}\right)}{\sqrt{\sum_{h=1}^{6}\left(y_i^{(h)} - \overline{Y_i}\right)^2} \times \sqrt{\sum_{h=1}^{6}\left(\hat{y}_i^{(h)} - \overline{\hat{Y}_i}\right)^2}} \tag{3}$$

where $Y_i \in \{y_i^{(h)}; h = 1, 2, \ldots, 6\}$ and $\hat{Y}_i \in \{\hat{y}_i^{(h)}; h = 1, 2, \ldots, 6\}$ represent the observations and the predictions of the NO_2 concentration at a vertical altitude of ith record.

We also used some other evaluation metrics including Spearman rank correlations, MAE, and RMSE

$$\rho_{\text{spearman}} = 1 - \left(6\sum_{i=1}^{n} d_i^2\right) / \left(n^3 - n\right) \tag{4}$$

$$MAE_h = \mathbb{E}\left\{\left|Y^{(h)} - \hat{Y}^{(h)}\right|\right\} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i^{(h)} - \hat{y}_i^{(h)}\right| \tag{5}$$

$$RMSE = \mathbb{E}\left\{\left(Y - \hat{Y}\right)^2\right\} = \mathbb{E}\left\{\mathbb{E}\{\left(Y^{(h)} - \hat{Y}^{(h)}\right)^2\right\} = \frac{1}{6}\sum_{h=1}^{6}\left\{\frac{1}{n}\sum_{i=1}^{n}\left(y_i^{(h)} - \hat{y}_i^{(h)}\right)^2\right\}. \tag{6}$$

IV. Experimental Results

In this section, we conducted four experiments to show the superior performance of MF-net over the WRF-CHEM (Section IV-A) model. Next, we demonstrated the unique applications of the MF-net (Section IV-B), the ability to predict NO_2 at night, and the more accurate evaluation of the transmissions between Beijing and the surrounding cities.

A. MF-Net Outperforms the WRF-CHEM Model

In this section, we first demonstrated that the MF-net outperformed the WRF-CHEM on the fourfold cross-validation of the MAX-DOAS-L data set. Then, we draw the same conclusion by comparing WRF-CHEM and the MF-net on the MAX-DOAS-S data set. By testing on the observations of 34 CNEMC stations and the satellite data of TROPOMI, we further demonstrated that the MF-net outperformed WRF-CHEM on the surface and VCDs prediction of NO_2, respectively.

1) 3-D Prediction of NO_2 on MAX-DOAS-L Data Set: The fourfold cross-validation experimental results are listed in Table III and Fig. 3.

From Table III, we could see that the MF-net outperforms the WRF-CHEM with smaller MAE, RMSE, and higher Spearman's rank correlation on all four sites. As shown in the last row of Table III, the average RMSE of MF-net was about $1.29 \times 10^{11} molec/cm^3$ lower and Spearman's rank correlation was 0.176 higher than those of the WRF-CHEM. Fig. 3 shows the superior performance of the MF-net over the WRF-CHEM more visually. The darker color bars represented the coefficients evaluated using the WRF-CHEM, and the lighter bars were evaluated using the MF-net. It was clear that for any altitude, the light bars were higher than the dark bars, demonstrating the better performance of the MF-net. Furthermore, the Pearson coefficients decreased with the increase of the altitude, suggesting that the MF-net perform better at lower altitude.

2) 3-D Prediction of NO_2 on MAX-DOAS-S Data Set: In this case, the MF-net was trained on all the observations in the MAX-DOAS-L data set and tested on the MAX-DOAS-S data set, and the results are shown in Table IV and Fig. 4.

From Table IV and Fig. 4, we could draw the same conclusion that the MF-net outperformed the WRF-CHEM with lower MAE and RMSE and higher Spearman's and

TABLE III
FOURFOLD CROSS-VALIDATION RESULTS OF WRF-CHEM AND MF-NET ON THE MAX-DOAS-L DATA SET

Tested	Method	MAE ($\times 10^{11} molec/cm^3$)						RMSE($\times 10^{11}$ $molec/cm^3$)	$\rho_{spearman}$
		0m	200m	400m	600m	800m	1000m		
GC	WRF-CHEM	1.36	1.25	1.08	0.94	0.78	0.60	1.54	0.528
	MF-net	**1.12**	**0.93**	**0.61**	**0.47**	**0.35**	**0.24**	**0.93**	**0.706**
GKD	WRF-CHEM	1.81	1.61	1.41	1.12	0.87	0.62	2.14	0.736
	MF-net	**1.40**	**1.10**	**0.67**	**0.50**	**0.36**	**0.25**	**1.06**	**0.778**
NC	WRF-CHEM	3.68	2.94	2.45	1.91	1.46	1.03	3.16	0.443
	MF-net	**1.22**	**1.13**	**0.62**	**0.46**	**0.32**	**0.25**	**1.13**	**0.747**
QKY	WRF-CHEM	3.34	2.72	2.24	1.88	1.54	1.10	3.09	0.535
	MF-net	**1.94**	**1.29**	**0.98**	**0.73**	**0.53**	**0.38**	**1.65**	**0.714**
4-fold	WRF-CHEM	2.55 ±0.98	2.13 ±0.72	1.79 ±0.57	1.46 ±0.44	1.16 ±0.34	0.84 ±0.23	2.48 ±0.68	0.560 ±0.108
	MF-net	**1.42 ±0.32**	**1.11 ±0.13**	**0.72 ±0.15**	**0.54 ±0.11**	**0.38 ±0.08**	**0.28 ±0.06**	**1.19 ±0.27**	**0.736 ±0.029**

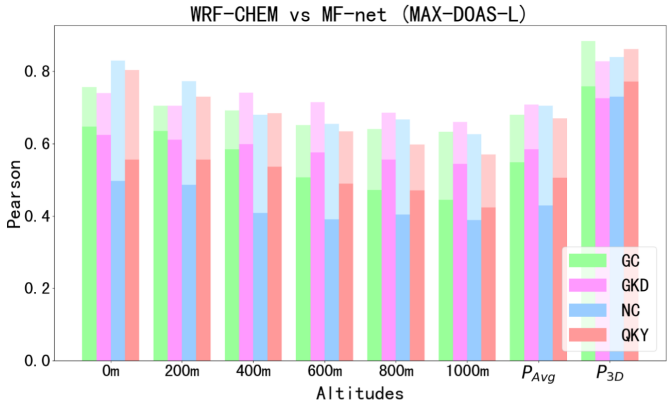


Fig. 3. Comparisons of the Pearson correlation coefficients of the WRF-CHEM and MF-net with the MAX-DOAS measurements as tested on the MAX-DOAS-L data set (fourfold cross-validation). The darker bars represent the performance of WRF-CHEM, while the lighter colors represent the portions that the MF-net exceeds the WRF-CHEM.

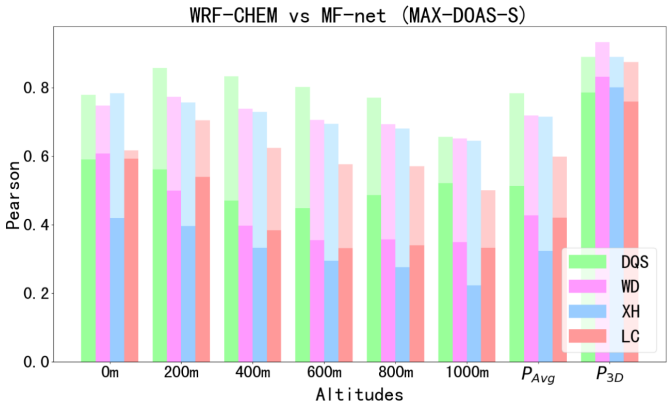


Fig. 4. Comparisons of the performance of the WRF-CHEM and MF-net as tested on MAX-DOAS-S data set.

Pearson correlations at most of the altitudes as tested on the sites of MAX-DOAS-S data set. And the average RMSE and Spearman's coefficients of MF-net are $1.01 \times 10^{11} molec/cm^3$ smaller and 0.215 higher than those of the WRF-CHEM, respectively. It was worth mentioning that, since the *LC* site was away from the center of the training sites, the performance of MF-net at the *LC* site was lower than at the other three sites.

3) Surface NO_2 Prediction as Tested on CNEMC Sites: To further verify the spatial generalization of MF-net, we tested MF-net and WRF-CHEM on the observation of 34 CNEMC stations which are more widely distributed over this region. As illustrated in Fig. 5, for any one of the 34 stations, the MF-net achieved higher Pearson correlation than WRF-CHEM. Majority of the Pearson coefficients evaluated using the MF-net were more than 0.1 higher than those evaluated using the WRF-CHEM model, suggesting that the MF-net has good spatial generalization property. From Fig. 5, we can see that there is a significant improvement at HuaYuan and YunGang stations and smaller improvements at Yanqing and Badaling.

4) VCD Predictions Evaluated on the Satellite Data Set: We compared the performance of the MF-net and WRF-CHEM on VCD of NO_2 within 1km of the ground using the satellite observations. The results are shown in the Fig. 6.

From Fig. 6, we can see that the MF-net outperforms the WRF-CHEM at most of the grid point inside of Beijing. However, we also found in some grids, in the northern mountain region, that the performance of MF-net was not good. It was because the stations of MAX-DOAS-L which we used to train the MF-net were mainly located in the south and the improvement of MF-net in the north is limited.

B. Applications of MF-Net

In this section, we show three applications of the proposed MF-net. First, by applying the MF-net, we extended the monitoring time of MAX-DOAS from 15 to 24 h including night. Second, we evaluated the NO_2 transmission at six different altitudes between Beijing and its neighboring cities from October 2016 to November 2019. By comparison, we found that the WRF-CHEM model overestimated the transmission between

TABLE IV

COMPARISON OF THE PERFORMANCE OF THE WRF-CHEM AND MF-NET AS TESTED ON MAX-DOAS-S DATA SET

Tested	Method	MAE ($\times 10^{11} molec/cm^3$)						RMSE($\times 10^{11}$ $molec/cm^3$)	$\rho_{spearman}$
		0m	200m	400m	600m	800m	1000m		
DQS	WRF-CHEM	2.82	2.54	1.97	1.19	0.71	0.49	2.60	0.626
	MF-net	**1.74**	**1.27**	**0.90**	**0.60**	**0.38**	**0.29**	**1.40**	**0.767**
WD	WRF-CHEM	1.26	1.17	0.89	0.74	0.60	0.45	1.26	0.433
	MF-net	**1.14**	**0.71**	**0.54**	**0.42**	**0.30**	**0.21**	**0.75**	**0.718**
XH	WRF-CHEM	4.27	3.23	2.17	1.58	1.21	0.88	3.10	0.320
	MF-net	**1.21**	**0.95**	**0.64**	**0.47**	**0.34**	**0.24**	**0.91**	**0.689**
LC	WRF-CHEM	**2.40**	**2.02**	1.69	1.26	0.97	0.70	2.08	0.560
	MF-net	2.91	2.31	**1.47**	**0.97**	**0.63**	**0.45**	**1.92**	**0.624**
Average	WRF-CHEM	2.69 $\pm$1.07	2.24 $\pm$0.75	1.68 $\pm$0.49	1.19 $\pm$0.30	0.87 $\pm$0.24	0.63 $\pm$0.17	2.26 $\pm$0.68	0.485 $\pm$0.118
	MF-net	**1.75 $\pm$0.71**	**1.31 $\pm$0.61**	**0.89 $\pm$0.36**	**0.61 $\pm$0.22**	**0.41 $\pm$0.13**	**0.30 $\pm$0.09**	**1.25 $\pm$0.46**	**0.700 $\pm$0.052**

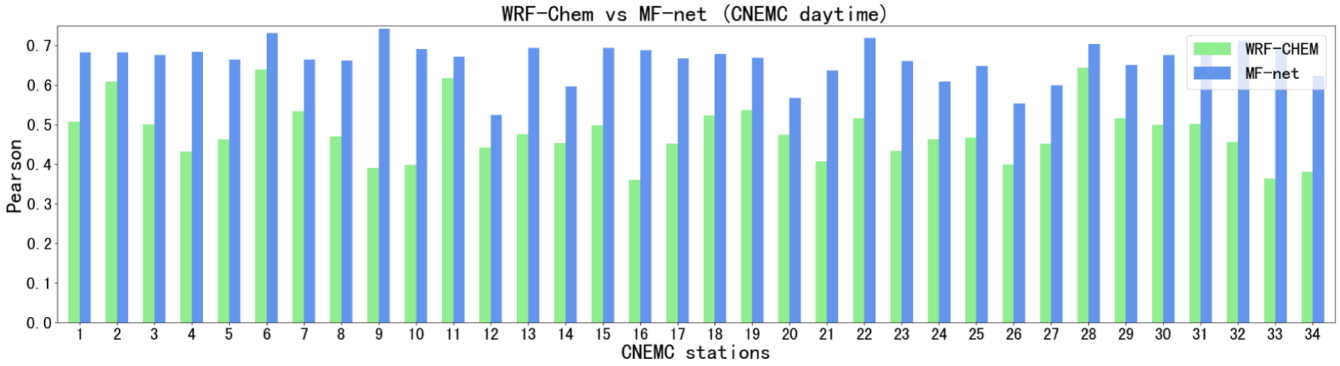


Fig. 5. Performance of the WRF-CHEM and MF-net as tested on the CNEMC observations during the daytime. The blue bars represent the performance of MF-net, while the green bars represent the performance of WRF-CHEM. The station numbers are given in Fig. 1.

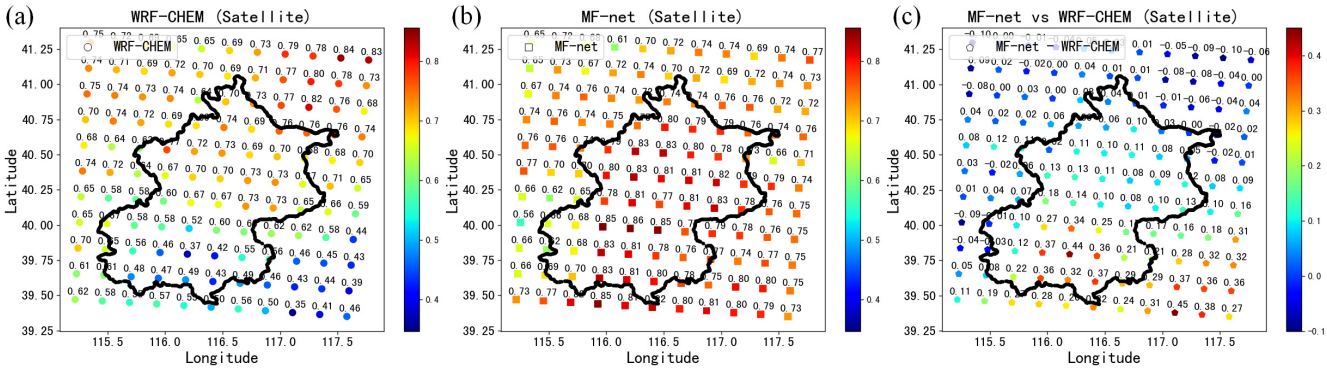


Fig. 6. Performance of WRF-CHEM and MF-net as tested on the observations of the satellite. (a) and (b) Pearson coefficient for NO_2 VCD prediction of the WRF-CHEM and MF-net, respectively. (c) Difference between the performance of MF-net and WRF-CHEM.

Beijing and Chengde and underestimated the transmission between Beijing and Zhangjiakou. Furthermore, we used the MF-net to monitor the distribution and transmission of a NO_2 pollution event in Beijing with a much higher temporal and spatial resolution.

1) MF-Net Extends the Monitoring Time of MAX-DOAS to the Whole Day: With the MF-net, we could predict the vertical profile of NO_2 even during night time, allowing us to extend the MAX-DOAS monitoring time from 15 to 24 h. To validate the predictions, we evaluated the Pearson coefficients with the 24-h observations from the CNEMC stations [Fig. 7(a)]. The Pearson coefficients were higher than 0.6 at majority of the CNEMC stations, suggesting the superior performance of the MF-net. Fig. 7(a) shows the comparisons of the WRF-CHEM and MF-net on 24-h observations at 34CNEMC stations. The results demonstrated that the MF-net outperform WRF-CHEM at all stations in the 24-h evaluation experiments. Fig. 7(b) and (c) shows the visual comparisons

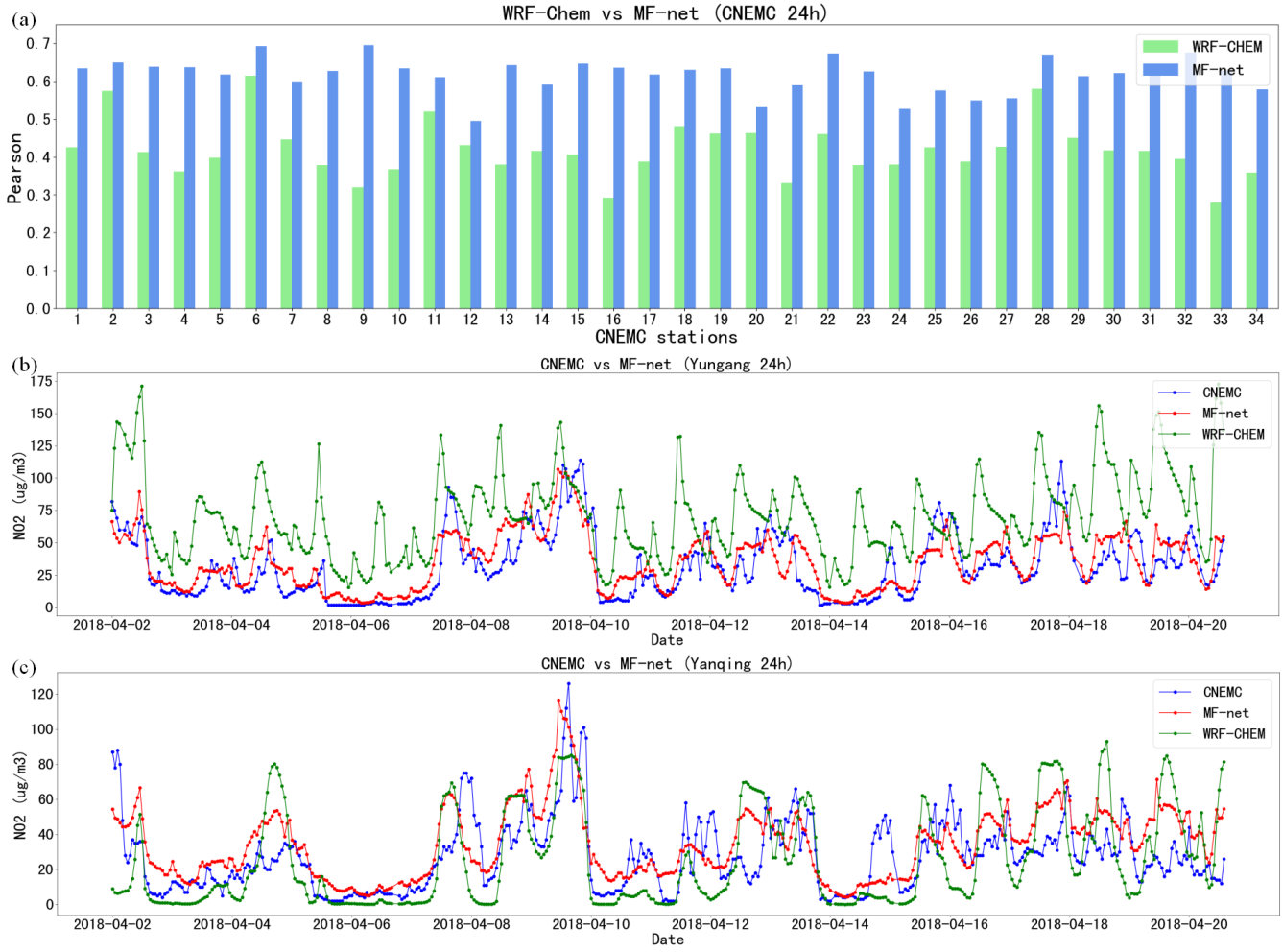


Fig. 7. Performance of the WRF-CHEM and MF-net as tested on the CNEMC observations during entire day (24 h). (a) Comparison of the Pearson coefficient of WRF-CHEM and MF-net evaluated on 34 CNEMC stations during the 24 h. (b) and (c) Visual interpolation results of the time series of MF-net, WRF-CHEM, and CNEMC observations from April 2, 2018 to April 20, 2018 at Yungang and Yanqing, respectively.

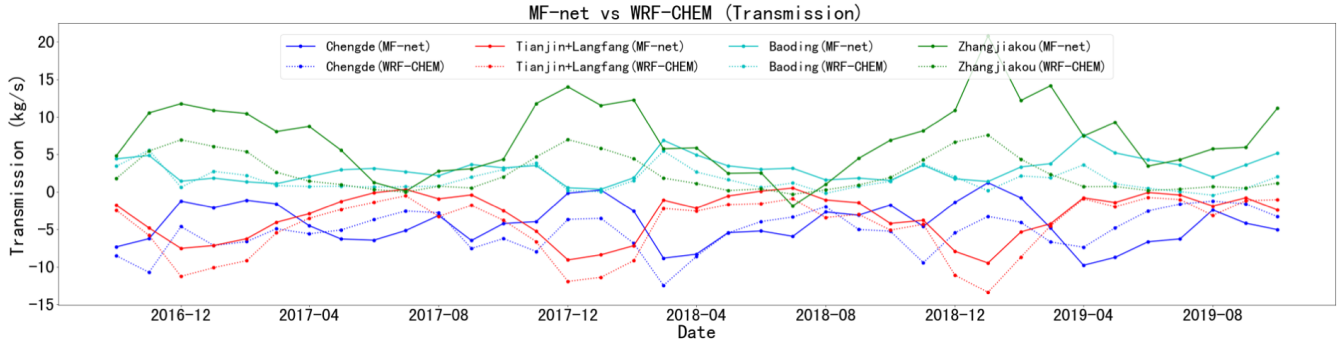


Fig. 8. Comparison of the WRF-CHEM and MF-net based NO_2 transmission between Beijing and its neighboring cities from October 2016 to November 2019. Comparison of WRF-CHEM and MF-net in calculating the monthly transmission of NO_2 between Beijing and its neighboring cities.

of the time series of MF-net and WRF-CHEM at two CNEMC stations where MF-net got the best and worst performance by testing on the observations of site, respectively. Compared with WRF-CHEM (green), the MF-net (red) is closer to the observations of CNEMC (blue).

2) More Accurate Transmission Evaluation Than WRF-CHEM: We compared the difference between WRF-CHEM and MF-net in statistic the transmission of NO_2 between Beijing and its neighboring cities [Fig. 8(a)] from October 2016 and November 2019.

As Fig. 8(b) shows the curve between Beijing and Baoding is always positive, indicating that the transport direction of NO_2 is from Baoding to Beijing. The same thing happened between Beijing and Zhangjiakou. On the contrary, NO_2 was usually transport from Beijing to Chengde, Tianjing, and Langfang. We can also see that Zhangjiakou accounts for the major cause of NO_2 pollution in Beijing, and Chengde plays a good role in the diffusion export of NO_2 in Beijing. One phenomenon worth mentioning is that the transmission volume curve between Beijing and Zhangjiakou is approximately

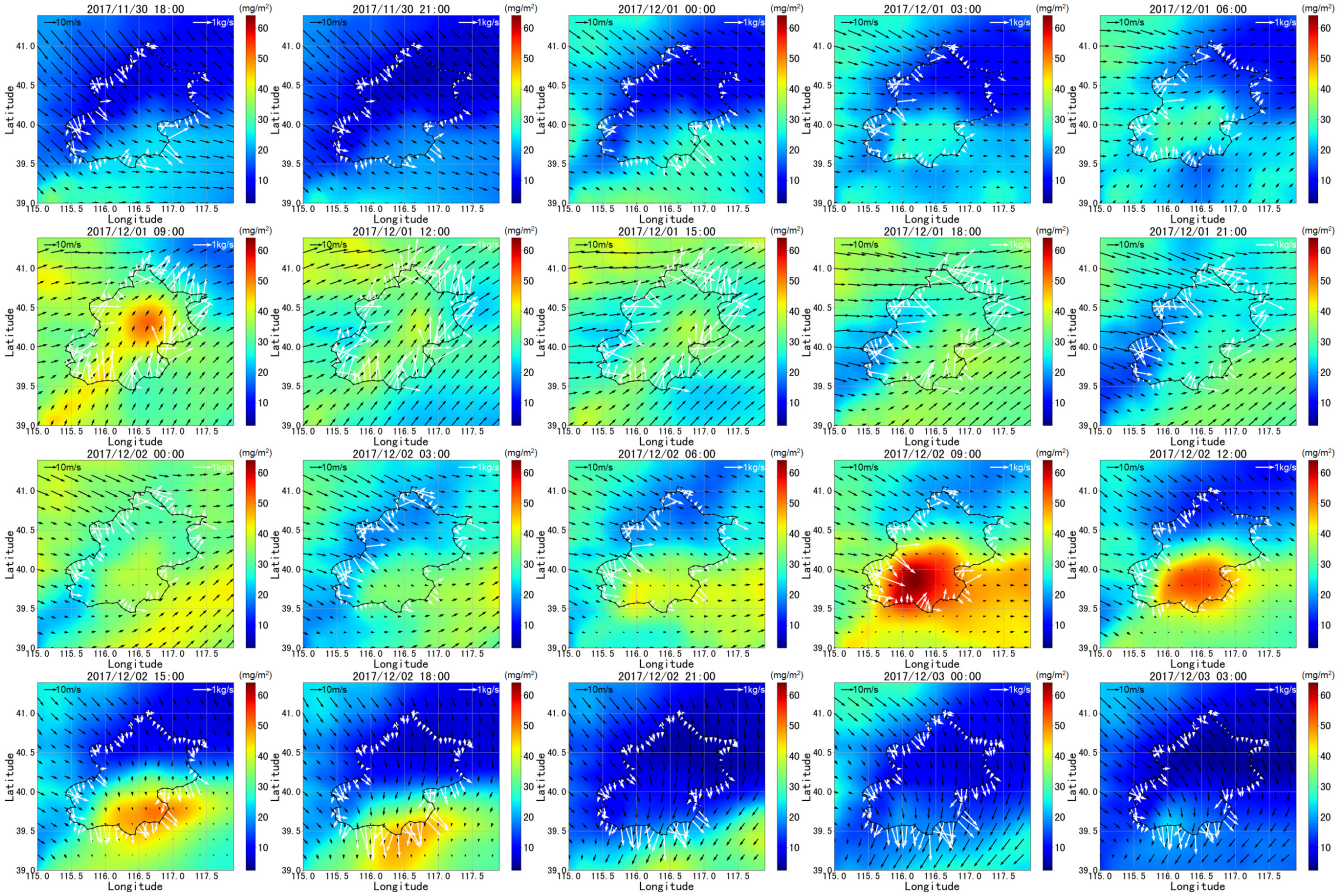


Fig. 9. NO_2 VCDs and transmission during the air pollution event in Beijing from November 30, 2017 to December 3, 2017 monitored by MF-net. The background color indicates the concentration of NO_2 VCDs (0 ∼ 1 km). The black arrows represent the wind and the white arrows show the NO_2 transmission along the border of Beijing.

symmetric about the X-axis with the transmission volume curve between Beijing and Tianjin Langfang. The explanation is that Zhangjiakou is in the west of Beijing, while the latter are in the south of Beijing.

In addition, we found that the biggest difference of transmission between MF-net based and WRF-CHEM based was at the border of Beijing and Chengde, as well as the Beijing and Zhangjiakou. Since we had verified the MF-net outperformed WRF-CHEM in Section IV-A, including testing on the satellite VCD observations at the border of Beijing (see Section IV-A4), the MF-net can provide better estimation of transmission evaluation.

3) Monitoring the Pollution Events in Continuous Space and Time: Since the MF-net has the capabilities of continuous monitoring on time and space, we used MF-net to monitor the distribution and transmission of NO_2 in pollution events with continuous time and space, and the results are shown in Fig. 9 and Table V.

Fig. 9 shows the distribution and transmission of NO_2 during an air pollution event in Beijing from November 30, 2017 to December 3, 2017. In the beginning, with a northwest wind, the NO_2 in Beijing and its surrounding areas were thin before the pollution came. From December 1, the wind direction gradually changed to the southwest, and NO_2 was transmitted to Beijing from Baoding at 9:00. Then, NO_2 started to accumulate southwest of Beijing from 6:00 to 12:00 on December 2, due to the reduced wind speed at 6:00. Finally, with the increase of the northwest wind speed, the pollution slowly disappeared.

Table V shows the assessment of intercity transport of NO_2 between Beijing and its neighboring cities in the pollution events. It is noted that there are NO_2 inflows and outflows through the administrative boundaries between Beijing and its adjacent cities. To distinguish them, we added "(+)" and "(−)" to represent the Beijing's NO_2 incoming and outgoing, respectively. As shown in Table V, the transmission between Beijing and Tianjin calculated by WRF-CHEM is almost equal as MF-net, similar as between Beijing and Baoding. But the transmission between Beijing and Chengde calculated by WRF-CHEM was more than MF-net. While for the transmission between Beijing and Zhangjiakou, the MF-net was higher than WRF-CHEM. To verify whether the MF-net exceeds WRF-CHEM in pollution events, we also tested the two methods on the satellite observations during the four-event periods. The average Pearson of MF-net at all grid points inside Beijing is 0.779, higher than 0.743 of WRF-CHEM, which showed superior performance of MF-net during the pollution event.

V. Discussion

The MF-net applies the meteorology data and the NO_2 vertical distribution from the WRF-CHEM simulation as the

TABLE V

STATISTICS OF THE NO_2 TRANSMISSION BETWEEN BEIJING AND ITS NEIGHBORING CITIES IN FOUR POLLUTION EVENTS USING WRF-CHEM AND MF-NET

Transport (kt)	Method	Chengde	Tianjin + Langfang	Baoding	Zhangjiakou
2017/11/30 ∼ 2017/12/04	WRF-CHEM	0.087(+), 2.810(-)	0.511(+), 3.406(-)	1.814(+), 0.911(-)	2.460(+), 0(-)
	MF-net	0.557(+), 1.461(-)	0.246(+), 2.615(-)	1.528(+), 0.751(-)	5.065(+), 0(-)
2018.01.12 ∼ 2018.01.21	WRF-CHEM	0.131(+), 8.995(-)	3.169(+), 9.692(-)	4.997(+), 1.492(-)	7.093(+), 0.651(-)
	MF-net	1.049(+), 5.578(-)	1.495(+), 6.251(-)	4.117(+), 0.816(-)	9.979(+), 0.995(-)
2018.11.23 ∼ 2018.11.28	WRF-CHEM	0.299(+), 4.151(-)	1.436(+), 5.027(-)	2.279(+), 1.140(-)	3.613(+), 0.324(-)
	MF-net	0.516(+), 2.629(-)	0.459(+), 3.243(-)	2.171(+), 0.649(-)	5.774(+), 0.329(-)
2018.12.19 ∼ 2018.12.22	WRF-CHEM	0.047(+), 2.463(-)	1.016(+), 3.366(-)	1.324(+), 0.314(-)	2.263(+), 0.351(-)
	MF-net	0.311(+), 1.560(-)	0.540(+), 2.232(-)	1.429(+), 0.201(-)	3.483(+), 0.542(-)

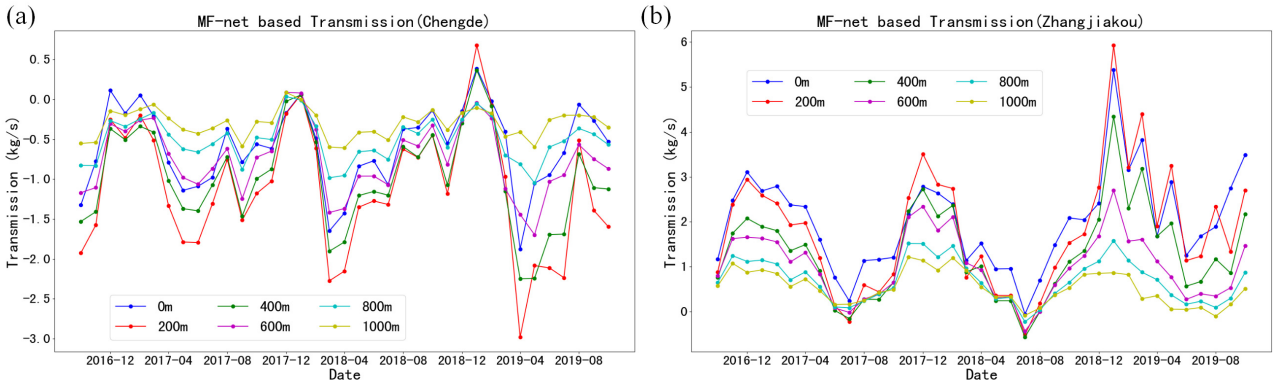


Fig. 10. Comparison of the different altitudes NO_2 transmission calculated by MF-net between Beijing and its neighboring cities from October 2016 to November 2019. (a) Different altitudes transmission of NO_2 between Beijing and Chengde. (b) Different altitudes transmission of NO_2 between Beijing and Zhangjiakou.

input, and the sparse but accurately measured NO_2 vertical distribution as the target. Through the training, the network establishes a high-dimensional mapping between the simulated data and the experimental observations. On the other hand, WRF-CHEM is based on a series of physical chemistry laws, requiring a sophisticated hyperparameter setting, which lacks adaptive capabilities to different environments. The MF-net gains the generalization property by training on measurements from various sites and allows us to predict the vertical NO_2 at locations without MAX-DOAS instruments. However, as shown in Table II, the further the areas are from the four sites used to train the network, the worse the predictions are. One possible explanation is that important factors that affect the NO_2 vertical distribution could be missing in the MF-net. For example, the landform or the local traffic pattern may affect the NO_2 vertical distribution and are different for areas further away from the training sites.

Besides the MF-net spatial generalization property, it also allows the predictions of vertical NO_2 during the nighttime, which lacks MAX-DOAS observations. To validate the predictions, we used the CNEMC 2-D surface NO_2 as the references. As shown in Fig. 7(a), the MF-net achieved a strong correlation with the measurements. Comparing with the WRF-CHEM model, the MF-net showed much better performance [Fig. 7(c)]. With the MF-net, therefore, we achieved the vertical prediction of NO_2 in range of $39.005 - 41.405$N and $115.005 - 117.905$E during the 24 h.

The performance of WRF-CHEM and MF-net decreased with the increase of the altitude (Tables III and IV). For WRF-CHEM, the main reason is that the observations of the assimilation inputs at high altitudes are few than those at low altitudes. For example, the meteorological measurements are scarce at high altitudes. The emission inventories, the vital information of WRF-CHEM, can only characterize the emission at the surface. Since the MF-net applied the WRF-CHEM outputs as the input, it was not surprising that its performance also decreased with an increase in the altitudes. Besides, the measurement error of the MAX-DOAS, used as the training label, also increases with height, which further decreases the accuracy of MF-net at high altitudes.

We applied both the WRF-CHEM and MF-net to study the NO_2 transmission at different altitudes between Beijing and its neighboring cities. We found significant differences at high altitudes between the two methods. However, since there are no direct NO_2 vertical measurements along the borders of Beijing current, we could not verify the two methods directly. Instead, we provided a brief comparison of WRF-CHEM and MF-net by comparing them with the satellite's VCDs observations, which demonstrated the better performance of the MF-net. We compared the different altitudes transmission of NO_2 calculated by MF-net as illustrated in the Appendix (Fig. 10). One interesting find was that the NO_2 transmission between Beijing and Chengde is mainly concentrated in the 200 m, while the transmission between Beijing and Zhangjiakou is mainly in the near-surface layer and decreased with the altitudes. A reasonable explanation is that the NO_2 concentration on the border between Beijing and Chengde is low, and the transmission at 200 m is large due to

TABLE VI

ABLATION EXPERIMENTS ON MAX-DOAS-L DATA SET TO ASSESS THE IMPORTANCE OF MF-NET INPUTS. THE ANNOTATION OF "MF-NET (I)" REPRESENTED THE MF-NET WAS ONLY TRAINED WITH NO_2 SIMULATION INPUT. THE "MF-NET (II)" MARKED THE NETWORK WAS TRAINED WITH INPUTTING ONLY THE FIVE METEOROLOGICAL VARIABLES WITHOUT THE NO_2 SIMULATION. THE "MF-NET" SHOWED THE RESULTS OF TRAINING WITH BOTH THE METEOROLOGICAL VARIABLES AND THE NO2 SIMULATION

Tested	Method	0 m	200 m	400 m	600 m	800 m	1000 m	Average
MAE ($\times 10^{11} molec/cm^3$)	WRF-CHEM	2.55 $\pm$0.98	2.13 $\pm$0.72	1.79 $\pm$0.57	1.46 $\pm$0.44	1.16 $\pm$0.34	0.84 $\pm$0.23	1.66 $\pm$0.54
	MF-net (i)	1.57 $\pm$0.25	1.20 $\pm$0.13	0.81 $\pm$0.09	0.61 $\pm$0.07	0.44 $\pm$0.05	0.32 $\pm$0.03	0.83 $\pm$0.10
	MF-net (ii)	1.51 $\pm$0.38	1.17 $\pm$0.15	0.76 $\pm$0.17	0.56 $\pm$0.13	0.40 $\pm$0.09	0.29 $\pm$0.06	0.78 $\pm$0.16
	MF-net	**1.42 $\pm$0.32**	**1.11 $\pm$0.13**	**0.72 $\pm$0.15**	**0.54 $\pm$0.11**	**0.38 $\pm$0.08**	**0.28 $\pm$0.06**	**0.74 $\pm$0.14**
RMSE ($\times 10^{11} molec/cm^3$)	WRF-CHEM	3.53 $\pm$1.13	3.11 $\pm$0.90	2.73 $\pm$0.74	2.27 $\pm$0.58	1.86 $\pm$0.45	0.41 $\pm$0.30	2.48 $\pm$0.68
	MF-net (i)	2.44 $\pm$0.53	1.87 $\pm$0.31	1.20 $\pm$0.21	0.92 $\pm$0.18	0.68 $\pm$0.14	0.50 $\pm$0.10	1.27 $\pm$0.23
	MF-net (ii)	2.46 $\pm$0.71	1.93 $\pm$0.38	1.16 $\pm$0.30	0.89 $\pm$0.25	0.65 $\pm$0.19	0.48 $\pm$0.13	1.26 $\pm$0.30
	MF-net	**2.29 $\pm$0.61**	**1.81 $\pm$0.33**	**1.10 $\pm$0.28**	**0.85 $\pm$0.23**	**0.63 $\pm$0.18**	**0.46 $\pm$0.13**	**1.19 $\pm$0.27**
Spearman's correlations	WRF	0.68 $\pm$0.06	0.65 $\pm$0.07	0.59 $\pm$0.10	0.55 $\pm$0.10	0.55 $\pm$0.10	0.53 $\pm$0.11	0.59 $\pm$0.09
	MF-net (i)	0.76 $\pm$0.04	0.71 $\pm$0.06	0.68 $\pm$0.07	0.65 $\pm$0.08	0.65 $\pm$0.08	0.63 $\pm$0.08	0.68 $\pm$0.07
	MF-net (ii)	0.78 $\pm$0.06	0.71 $\pm$0.04	0.68 $\pm$0.03	0.66 $\pm$0.03	0.66 $\pm$0.03	0.63 $\pm$0.02	0.69 $\pm$0.03
	MF-net	**0.83 $\pm$0.03**	**0.76 $\pm$0.03**	**0.73 $\pm$0.04**	**0.71 $\pm$0.04**	**0.70 $\pm$0.04**	**0.69 $\pm$0.04**	**0.74 $\pm$0.03**
Pearson correlation	WRF	0.58 $\pm$0.07	0.57 $\pm$0.07	0.53 $\pm$0.09	0.49 $\pm$0.08	0.48 $\pm$0.06	0.45 $\pm$0.07	0.52 $\pm$0.07
	MF-net (i)	0.73 $\pm$0.01	0.69 $\pm$0.01	0.64 $\pm$0.05	0.61 $\pm$0.05	0.59 $\pm$0.04	0.57 $\pm$0.04	0.64 $\pm$0.03
	MF-net (ii)	0.73 $\pm$0.07	0.68 $\pm$0.05	0.65 $\pm$0.03	0.61 $\pm$0.03	0.60 $\pm$0.03	0.57 $\pm$0.03	0.64 $\pm$0.03
	MF-net	**0.78 $\pm$0.04**	**0.73 $\pm$0.03**	**0.70 $\pm$0.03**	**0.66 $\pm$0.04**	**0.65 $\pm$0.04**	**0.62 $\pm$0.04**	**0.69 $\pm$0.02**

large wind speed at high altitude. While the NO_2 concentration on the boundary between Beijing and Zhangjiakou was high, the high concentration of surface NO_2 caused larger transmission at lower altitudes.

In Section IV-B3, we proved the better performance of MF-net during the pollution events using the satellite observations, and demonstrated a higher transmission between Beijing and Zhangjiakou, a less transmission between Beijing and Chengde. As the previous air pollution research paid more attention to the impact of Baoding on Beijing, we found the transmission from Zhangjiakou to Beijing dominated the NO_2 pollution during the pollution events. Our results provided scientific data analysis for the pollution control policy making.

Currently, we only had four MAX-DOAS observations sites with limited data, which prevented us from applying deeper neural networks. Furthermore, the stations were within a local region and lacked the diversity of different landforms, land uses, and traffic patterns. Therefore, the MF-net did not include these factors, which limited the application of the network to locations further away from the training sites. By extending the coverage areas of the MAX-DOAS, we expect to improve the spatial generalization property of the MF-net in the future. The MF-net is not limited to the study of NO_2, it could be used to predict the 3-D vertical profiles of other pollutants, such as aerosol and so on.

VI. CONCLUSION

The 3-D prediction of NO_2 is crucial in understanding the pollution mechanisms, tracking pollutant transport at different altitudes, and formulating the pollution prevention and control measures. In this article, we developed an MF-net to achieve 3-D prediction of NO_2 within Beijing and its surrounding areas ($115.005 - 117.905$E and $39.005 - 41.405$N). The MF-net built a map from the WRF-CHEM simulation to MAX-DOAS observations and showed superior performance in experiment results. Compared with WRF-CHEM, the MF-net provided a more accurate tool to estimate the 3-D distribution and transmission of NO_2 with high spatial-temporal continuity. One interesting find was that previous WRF-CHEM significantly underestimated the transfer of NO_2 from Zhangjiakou to Beijing. The study could help us to better understand the

NO_2 pollution in Beijing and provide a scientific basis for pollution control policies.

APPENDIX

To assess the importance of MF-net input including meteorological variables and NO_2 simulation, we did the ablation experiments on the MAX-DOAS-L data set with three different experimental conditions. In the first experiment, the MF-net was trained only with the simulated NO_2 as the input, without using the meteorological variables. The results are marked with "MF-net (i)" as shown in Table VI in the Appendix. In the second experiment, we trained the MF-net only with the meteorological variables as the input, without using the simulated NO_2. The results were marked with "MF-net (ii)." The third experiment condition was training the MF-net with both simulated NO_2 and meteorological variables marked with "MF-net." Since the average indicators were a more comprehensive index to evaluate the performance of these methods, we analyzed the results by comparing the average indicators. As shown in Table VI, both "MF-net (i)" and "MF-net (ii)" achieved lower average MAE and RMSE, higher Spearman's and Pearson correlation significantly outperform the WRF-CHEM. And the "MF-net" achieved the lowest average MAE and RMSE and the highest Spearman's and Pearson correlation, suggesting the best performance of the training of the MF-net using both simulated NO_2 and meteorological variables. It was because the two factors containing the complementary information of NO_2.

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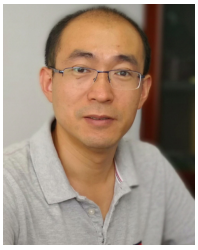
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